Mohammad Abu Hijleh

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Professional Summary

Computer Engineering student with a strong foundation in machine learning, computer vision, and software engineering. Experienced developing and deploying AI-powered systems using Python, FastAPI, and REST APIs. Interested in building scalable AI applications integrating machine learning with backend services, with a growing focus on Generative AI and intelligent systems.

Technical Skills

Programming: Python, Java, SQL, C

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

AI Domains: Machine Learning, Computer Vision, Deep Learning, Generative AI, Retrieval-Augmented Generation (RAG)

LLM & GenAI Tools: LangChain, FAISS, Ollama, vector databases

Backend & Deployment: FastAPI, Flask, REST APIs, Docker, MySQL, PostgreSQL, Streamlit | Tools: Git, Linux (Ubuntu)

AI & Machine Learning Projects

AI Document Assistant (RAG) — *Python, LangChain, FAISS, FastAPI, Streamlit* — *Code* *2026*

- Built an end-to-end RAG system enabling natural language Q&A over user-uploaded documents, running fully locally with no external API dependencies.
- Implemented a document ingestion pipeline with parsing, chunking, and embedding generation using LangChain.
- Developed semantic retrieval using FAISS vector search and `nomic-embed-text` embeddings to retrieve relevant document passages.
- Integrated a local LLM (qwen3:4b via Ollama) and deployed the system through a FastAPI REST API with a Streamlit chat interface.

Text-to-Video Retrieval System — *Python, Deep Learning, Flask* — *Code* *2026*

- Multimodal text–video retrieval system on MSR-VTT (10,000 videos, 200,000 captions).
- Implemented feature extraction, semantic similarity matching, and ranking for query-to-video retrieval.
- Achieved **Recall@1: 19.6%**, **Recall@5: 47.1%**, and **Recall@10: 60.5%**.
- Reached **mAP: 0.328** and deployed the system via a Flask REST API for real-time queries.

Handwritten Arabic Digit Recognition — *Python, CNN, Computer Vision, FastAPI* — *Code* *2026*

- CNN-based handwritten Arabic digit classification system (0–9) with ˜157K parameters.
- Built preprocessing and training pipelines on a 9,350-image dataset with a 1,403-image test split.
- Achieved **98.86%** test accuracy with only **16** misclassified samples.
- Deployed the trained model with FastAPI for real-time inference.

Education

Birzeit University **Birzeit, Palestine**
Bachelor of Computer Engineering, **GPA:** *84.9* *Sep 2022–Feb 2027*
Relevant Coursework: Machine Learning, Computer Vision, Data Structures & Algorithms, Data Base, Computer Networks.

Certifications & Courses

- Machine Learning with TensorFlow Nanodegree — Udacity — Certificate — Sep 2025–Jan 2026
- Programming for Data Science with Python Nanodegree — Udacity — Feb 2026–Present
- DataCamp Program — Gaza Sky Geeks (Mercy Corps) — Sep 2025–Sep 2026
Competitive one-year data science program sponsored by an international NGO.
- English Access Microscholarship — U.S. Embassy (AMIDEAST) — Certificate — 2018–2020
Two-year U.S. Embassy-sponsored program focused on English proficiency and leadership.